

Dual Conic Proxy for Semidefinite Relaxation of AC Optimal Power Flow

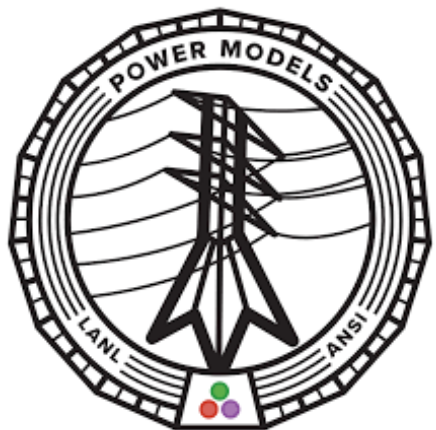


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AC-OPF (Optimal Power Flow)



MATPOWER

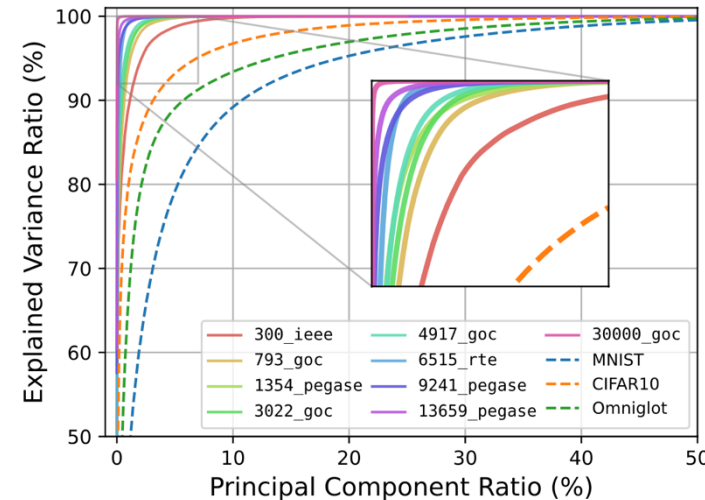
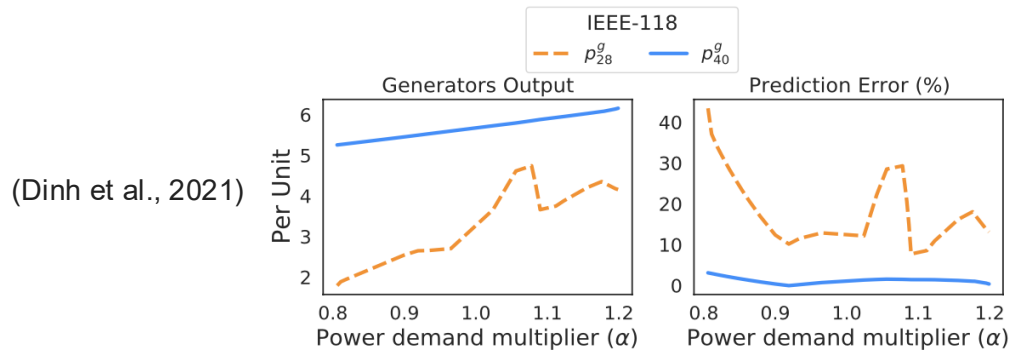


$$\begin{aligned}
 & \min \sum_{i \in \mathcal{N}} c_i p_i^g \\
 & \text{s.t.} \quad \mathbf{S}_i^g - \sum_{e \in \mathcal{E}_i^+} \vec{\mathbf{S}}_e - \sum_{e \in \mathcal{E}_i^-} \tilde{\mathbf{S}}_e - (Y_i^s)^* |\mathbf{V}_i|^2 = \boxed{\mathbf{S}_i^d} \quad \forall i \in \mathcal{N} \quad \begin{array}{l} \text{Load (power demand)} \\ \text{Supply-demand balance} \end{array} \\
 & \quad \vec{\mathbf{S}}_{ij} = (Y_{ij} + Y_{ij}^c)^* \frac{|\mathbf{V}_i|^2}{|T_{ij}|^2} - Y_{ij}^* \frac{\mathbf{V}_i \mathbf{V}_j^*}{T_{ij}} \quad \forall ij \in \mathcal{E} \\
 & \quad \tilde{\mathbf{S}}_{ij} = (Y_{ij} + Y_{ji}^c)^* |\mathbf{V}_j|^2 - Y_{ij}^* \frac{\mathbf{V}_i^* \mathbf{V}_j}{T_{ij}^*} \quad \forall ij \in \mathcal{E} \quad \left. \begin{array}{l} \\ \end{array} \right\} \text{Power flow equations} \\
 & \quad |\vec{\mathbf{S}}_{ij}|, |\tilde{\mathbf{S}}_{ij}| \leq \bar{s}_{ij} \quad \forall ij \in \mathcal{E} \quad \text{Line thermal limit} \\
 & \quad \underline{v}_i \leq |\mathbf{V}_i| \leq \bar{v}_i \quad \forall i \in \mathcal{N} \quad \text{Voltage (complex)} \\
 & \quad \underline{\theta}_{ij} \leq \boxed{\theta_i} - \boxed{\theta_j} \leq \bar{\theta}_{ij} \quad \forall ij \in \mathcal{E} \\
 & \quad \underline{p}_i^g \leq \boxed{p_i^g} \leq \bar{p}_i^g \quad \forall i \in \mathcal{N} \quad \text{Active (real) power generation} \\
 & \quad \underline{q}_i^g \leq \boxed{q_i^g} \leq \bar{q}_i^g \quad \forall i \in \mathcal{N} \quad \text{Reactive (imaginary) power generation}
 \end{aligned}$$



Machine Learning Proxy for OPF: Why?

- ▶ Relatively easy to learn...

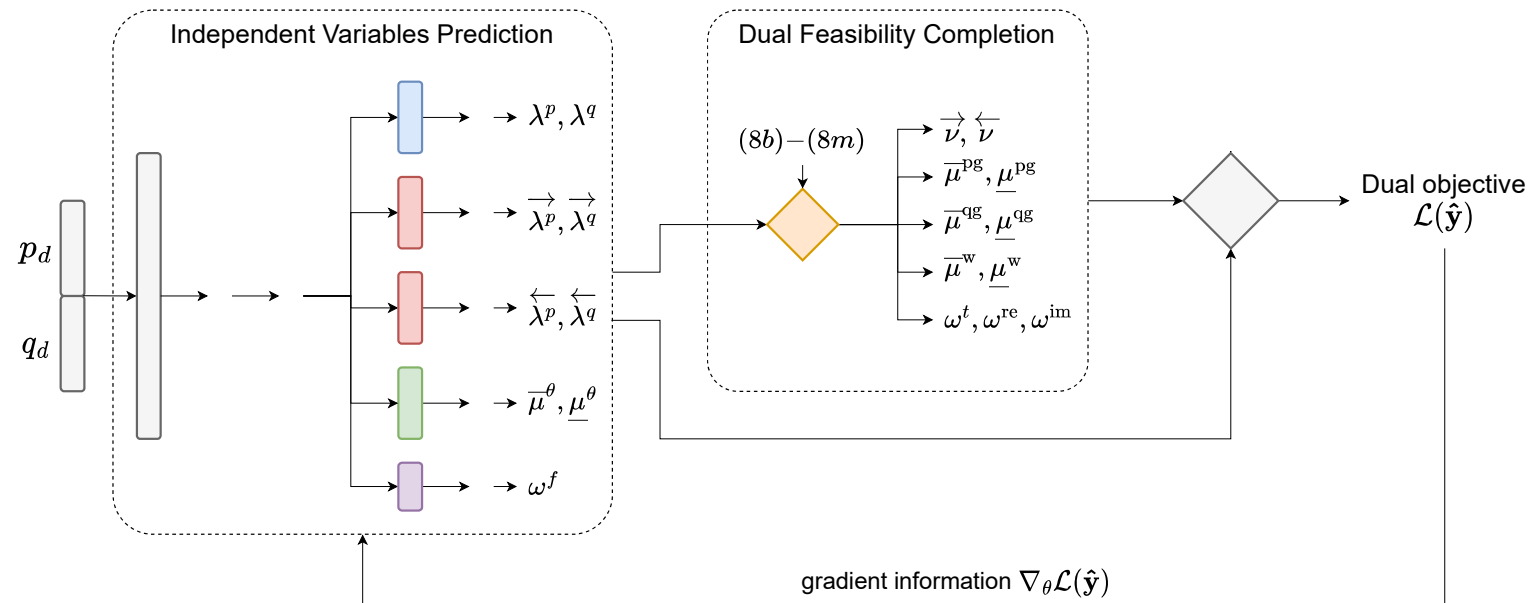


(Park et al., 2023)

- ▶ ...but with significant potential, most notably by transferring online overhead to offline
 - Real-time risk assessment (simulation)
 - Real-time economical operations in the age of renewable energy
 - Tractable expansion planning
- ▶ e.g. max violation: 0.36 p.u., mean violation: $\sim 10^{-4}$ p.u., mean optimality gap: 0.04% on a 2869-bus system (Chen et al., 2025)

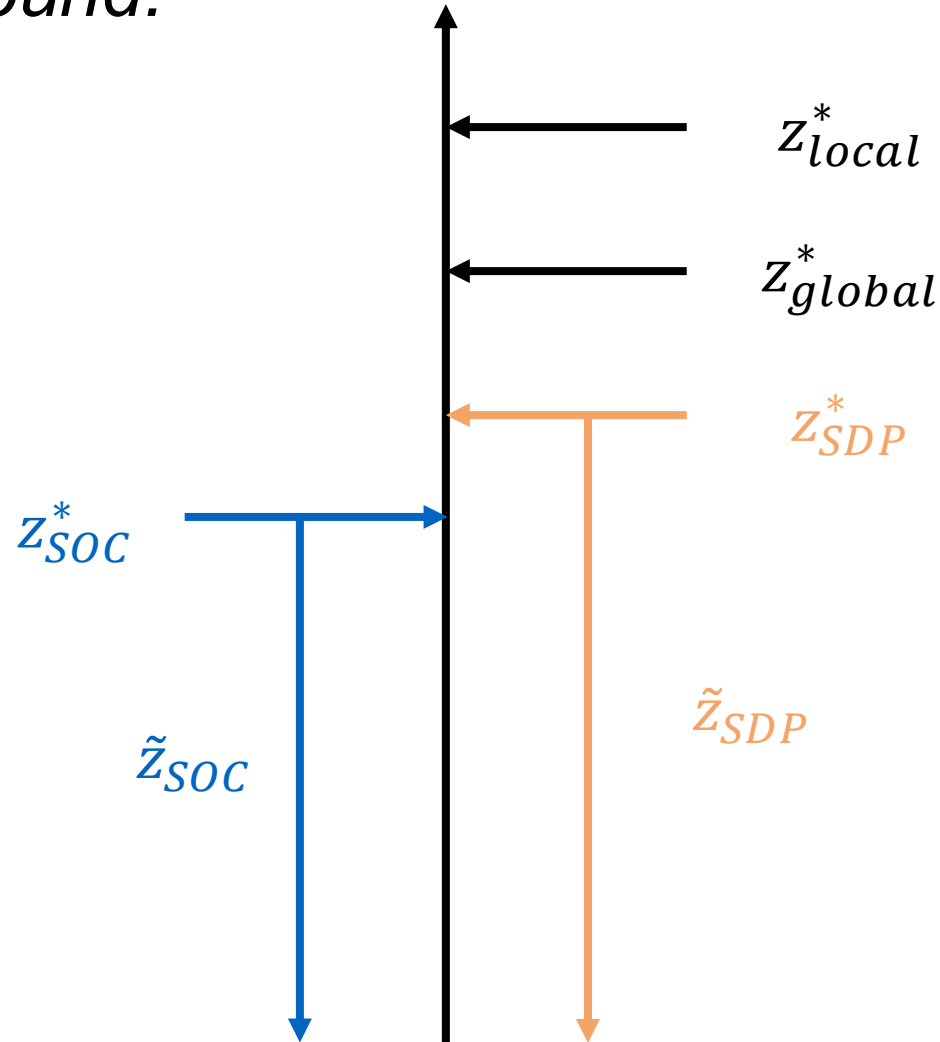
Machine Learning Proxy for OPF: Caveat

- ▶ No certificates for optimality
 - Usually, an interior point method solver can provide a lower bound
- ▶ An obstacle for industrial acceptance
- ▶ The solution: dual conic proxy (DCP): an ML proxy (Qiu et al., 2024) for certifying AC-OPF solutions (like solvers) efficiently



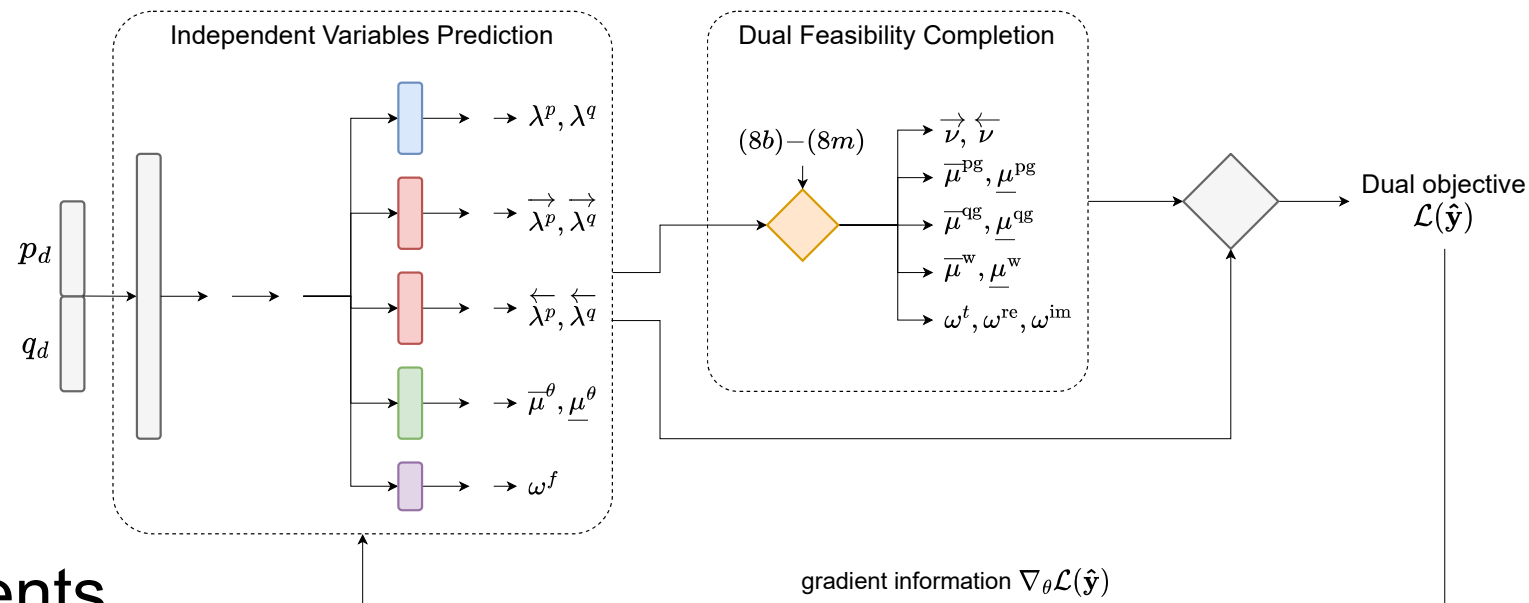
Convex Relaxation Dual Certificate

DCP outputs a feasible solution to the dual of a convex relaxation, yielding a valid lower bound.



The Dual Conic Proxy (DCP)

- ▶ A machine learning proxy that produces feasible solution to the dual of a convex problem.



- ▶ Key ingredients

- Variable upper/lower bounds in the primal $\Rightarrow$ slack variables in the dual.
- Identifying dependent dual variables and completing them optimally (filling in the blanks).
- Using the dual objective function as training loss.

Recovery Using Partial Solution

Model 3 The DSDP-OPF model

$$\max_{\lambda, \mu, \mathbf{v}} \sum_{i \in \mathcal{N}} \left(p_i^d \lambda_i^p + q_i^d \lambda_i^q + \underline{v}_i^2 \underline{\mu}_i^w - \bar{v}_i^2 \bar{\mu}_i^w + \underline{p}_i^g \underline{\mu}_i^{pg} - \bar{p}_i^g \bar{\mu}_i^{pg} + \underline{q}_i^g \underline{\mu}_i^{qg} - \bar{q}_i^g \bar{\mu}_i^{qg} \right) - \sum_{e \in \mathcal{E}} \bar{s}_e (\bar{\mathbf{v}}_e^s + \tilde{\mathbf{v}}_e^s) \quad (6a)$$

$$\text{s.t.} \quad \boxed{\lambda_i^p} + \boxed{\underline{\mu}_i^{pg}} - \boxed{\bar{\mu}_i^{pg}} = c_i \quad \forall i \in \mathcal{N} \quad (6b)$$

$$\boxed{\lambda_i^q} + \boxed{\underline{\mu}_i^{qg}} - \boxed{\bar{\mu}_i^{qg}} = 0 \quad \forall i \in \mathcal{N} \quad (6c)$$

$$-\boxed{\lambda_i^p} - \boxed{\bar{\lambda}_{ij}^p} + \boxed{\bar{\mathbf{v}}_{ij}^p} = 0 \quad \forall ij \in \mathcal{E} \quad (6d)$$

$$-\boxed{\lambda_i^q} - \boxed{\bar{\lambda}_{ij}^q} + \boxed{\bar{\mathbf{v}}_{ij}^q} = 0 \quad \forall ij \in \mathcal{E} \quad (6e)$$

$$-\boxed{\lambda_j^p} - \boxed{\bar{\lambda}_{ij}^p} + \boxed{\bar{\mathbf{v}}_{ij}^p} = 0 \quad \forall ij \in \mathcal{E} \quad (6f)$$

$$-\boxed{\lambda_j^q} - \boxed{\bar{\lambda}_{ij}^q} + \boxed{\bar{\mathbf{v}}_{ij}^q} = 0 \quad \forall ij \in \mathcal{E} \quad (6g)$$

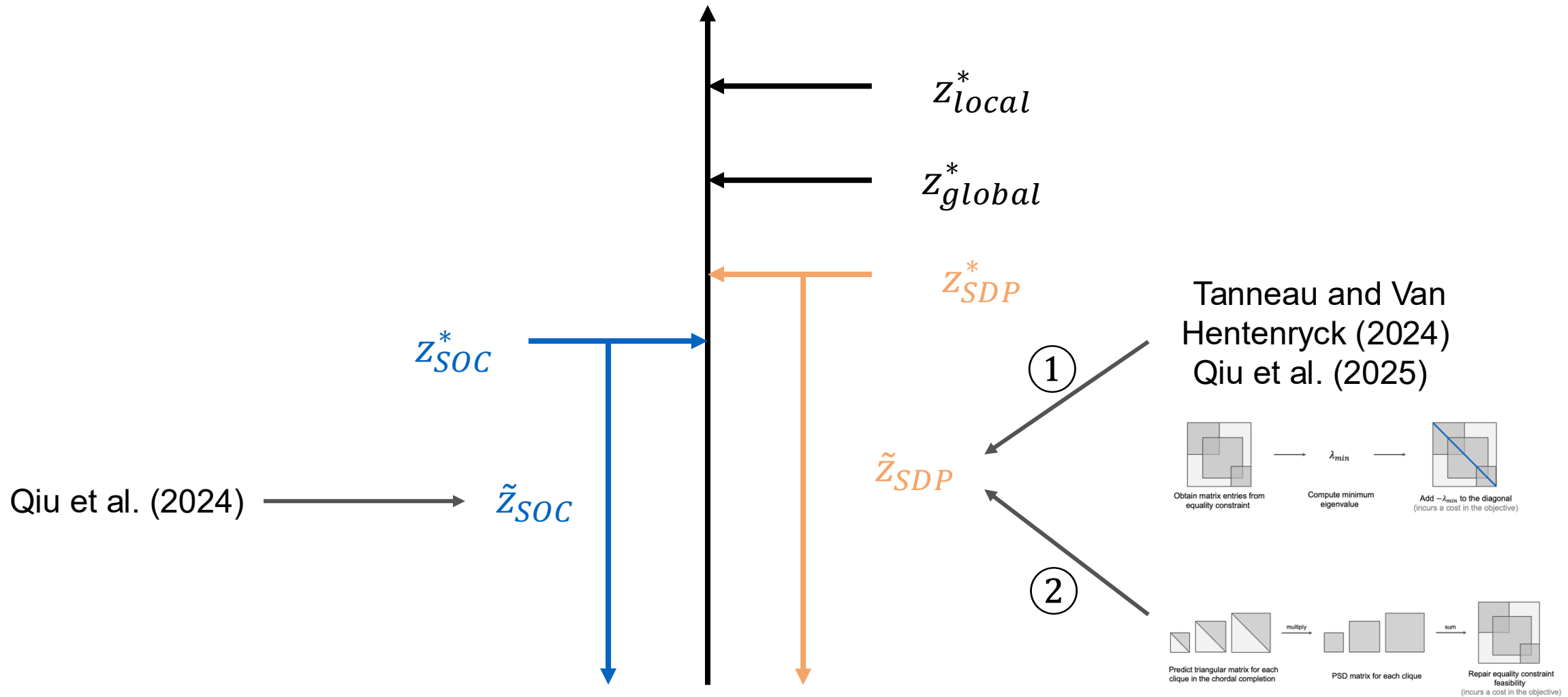
$$\boxed{\mathcal{A}_R(\lambda, \mu, \mathbf{v})} + \boxed{\mathbf{j} \mathcal{A}_I(\lambda, \mu, \mathbf{v})} + \mathbf{S} = 0 \quad (6h)$$

$$\underline{\mu}^{pg}, \bar{\mu}^{pg}, \underline{\mu}^{qg}, \bar{\mu}^{qg}, \underline{\mu}^w, \bar{\mu}^w \geq 0 \quad (6i)$$

$$\bar{\mathbf{v}}_{ij} = (\boxed{\bar{\mathbf{v}}_{ij}^s}, \boxed{\bar{\mathbf{v}}_{ij}^p}, \boxed{\bar{\mathbf{v}}_{ij}^q}) \in Q^3, \tilde{\mathbf{v}}_{ij} = (\boxed{\tilde{\mathbf{v}}_{ij}^s}, \boxed{\tilde{\mathbf{v}}_{ij}^p}, \boxed{\tilde{\mathbf{v}}_{ij}^q}) \in Q^3 \quad \forall ij \in \mathcal{E} \quad (6j)$$

$$\mathbf{S} \in \mathbb{H}_+^{|\mathcal{N}|} \quad (6k)$$

Convex Relaxation Dual Certificate

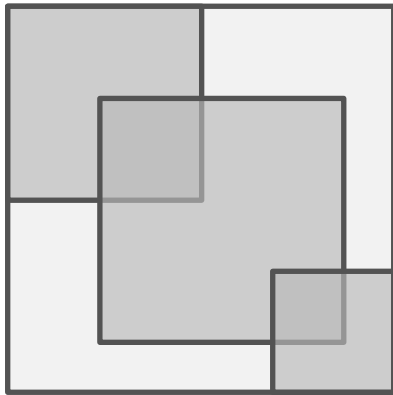


Qiu, G., Tanneau, M., & Van Hentenryck, P. (2024). Dual conic proxies for ac optimal power flow. *Electric Power Systems Research*, 236, 110661.

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SDP Method 1 (*SDP_Eigen*)

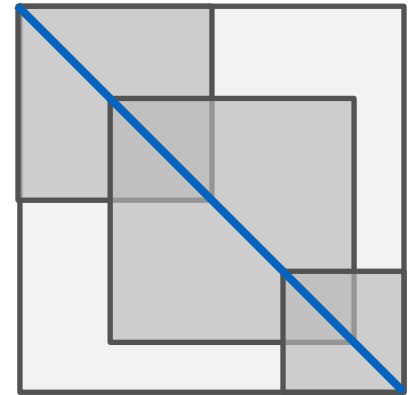


Recover matrix entries from
equality constraint



λ_{min}

Compute minimum
eigenvalue



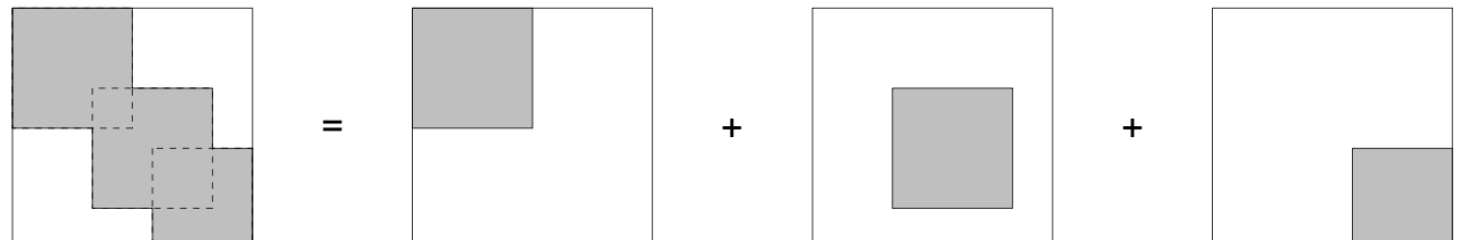
Add $-\lambda_{min}$ to the diagonal
(incurs a cost in the objective)

Positive semidefinite matrices with chordal sparsity pattern

$S \in \mathbf{S}_E^n$ is positive semidefinite if and only if it can be expressed as

$$S = \sum_{\text{cliques } \gamma_i} P_{\gamma_i}^T H_i P_{\gamma_i} \quad \text{with } H_i \geq 0$$

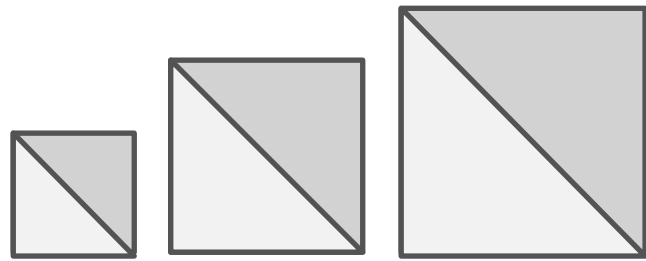
(for an index set β , P_β is 0-1 matrix of size $|\beta| \times n$ with $P_\beta x = x_\beta$ for all x)



$$S \geq 0 \quad = \quad P_{\gamma_1}^T H_1 P_{\gamma_1} \geq 0 \quad + \quad P_{\gamma_2}^T H_2 P_{\gamma_2} \geq 0 \quad + \quad P_{\gamma_3}^T H_3 P_{\gamma_3} \geq 0$$

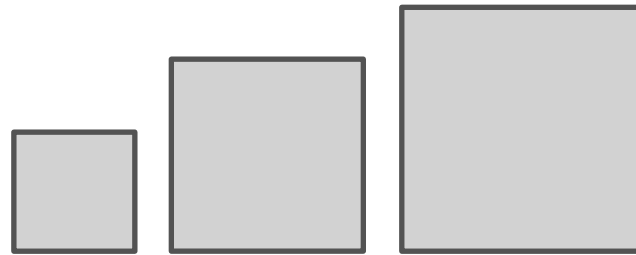
[Griewank and Toint 1984] [Agler, Helton, McCullough, Rodman 1988]

SDP Method 2 (*SDP_Clique*)



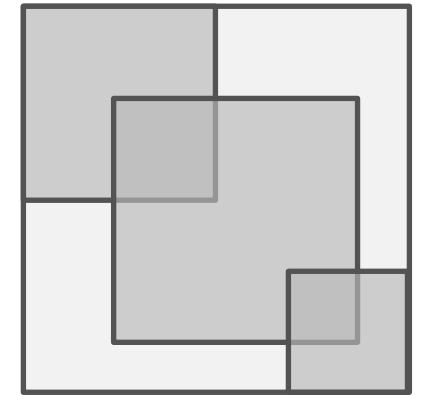
Predict triangular matrix for each
clique in the chordal completion

multiply →



PSD matrix for each clique

sum →



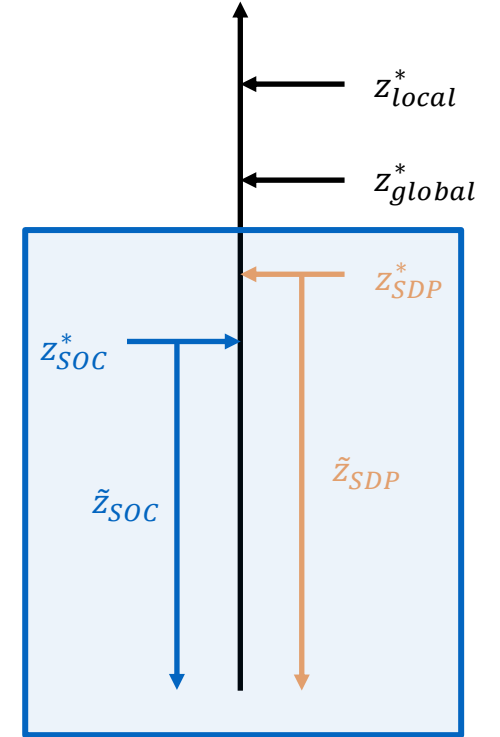
Repair equality constraint
feasibility
(incurs a cost in the objective)

Experiment Results

DCP PERFORMANCE RESULTS: % GAP W.R.T. z_{SDP}^*

System	Mosek	DCP		
	z_{SOC}^*	$\hat{z}_{SOC}$	$\hat{z}_{SDP_Eigen}$	$\hat{z}_{SDP_Clique}$
ieee14	0.093 (0.081)	0.244 (4.861)	0.723 (4.960)	0.438 (4.990)
ieee118	0.791 (0.569)	0.913 (0.697)	0.778 (0.584)	1.274 (1.205)
ieee300	1.183 (0.606)	3.285 (0.702)	4.741 (0.896)	5.577 (1.448)

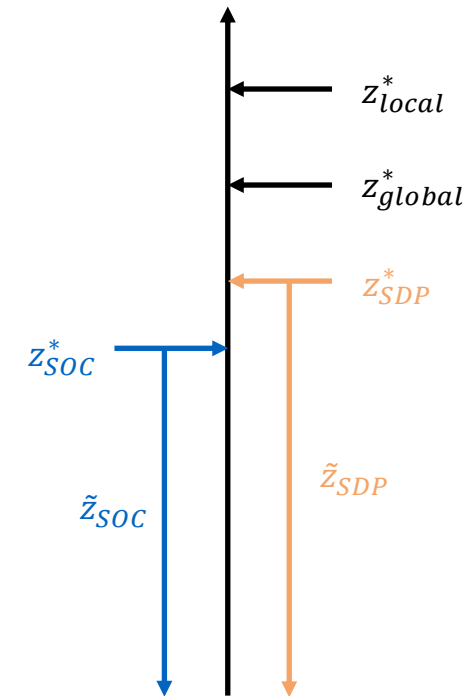
Statistics presented are geometric mean (standard deviation). Numbers in system names indicate numbers of buses.



- ▶ Data sampled using PGLearn (Klamkin et al., 2025).
- ▶ On *ieee118*, the *SDP_Eigen* proxy achieves theoretically better bounds than the *SOC* proxy on average.
- ▶ *On *pegase1354* (not shown), *SDP_Eigen* incurs significant overhead in training (> 1 week); *SDP_Clique* achieves poor optimality gaps.

Conclusion

- ▶ Dual conic proxies (DCP) complement AC OPF proxies to certify how optimal solution predictions are, paving the path to industrial adoption.
- ▶ Development of DCP on the stronger SDP relaxation requires more effort.
- ▶ Future directions include more efficient minimum eigenvalue algorithm exploiting sparsity and batched multiplication.



- ▶ Chen, K., Bose, S., & Zhang, Y. (2025). Physics-Informed Gradient Estimation for Accelerating Deep Learning-Based AC-OPF. *IEEE Transactions on Industrial Informatics*.
- ▶ Dinh, M. H., Fioretto, F., Mohammadian, M., & Baker, K. (2021). Towards understanding the unreasonable effectiveness of learning AC-OPF solutions. *arXiv preprint arXiv:2111.11168*.
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- ▶ Park, S., Chen, W., Mak, T. W., & Van Hentenryck, P. (2023). Compact optimization learning for AC optimal power flow. *IEEE Transactions on Power Systems*, 39(2), 4350-4359.
- ▶ Qiu, G., Tanneau, M., & Van Hentenryck, P. (2024). Dual conic proxies for AC optimal power flow. *Electric Power Systems Research*, 236, 110661.
- ▶ Tanneau, M., & Van Hentenryck, P. (2024). Dual lagrangian learning for conic optimization. *arXiv preprint arXiv:2402.03086*.